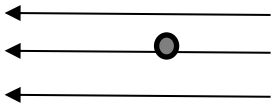
Practice Problems
Calculus-Based Physics II

Name: _____ **Date:** _____

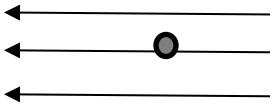
Chapter 36 Part A Electric Fields

1) An electric field is pointing to the left, an object with a net positive charge is placed in this field, what direction does the object experience a force?



A) To the left.
B) To the right.
C) It does not experience a force.
D) Not enough information is given to know what direction.

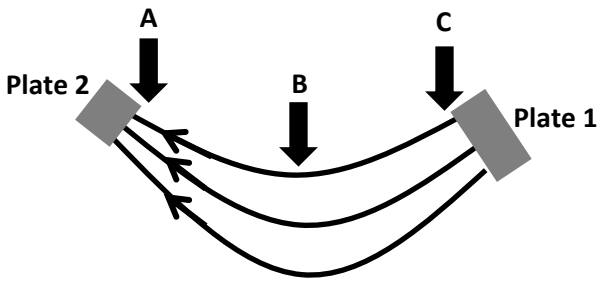
2) An electric field is pointing to the left, an object with a net negative charge is placed in this field, what direction does the object experience a force?



A) To the left.
B) To the right.
C) It does not experience a force.
D) Not enough information is given to know what direction.

3) An electric field is in a space between plate 1 and plate 2 in the direction from right to left, which section of the electric field indicated by the letters (A, B or C) has the largest field strength?

A) Section A is largest
B) Section B is largest
C) Section C is largest
D) They are all the same field strength



4) Using the diagram above, what must the charge on Plate 2 be in order to create this electric field?

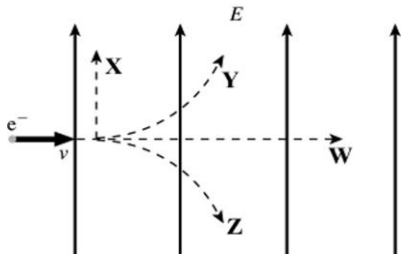
A) Positive
B) Negative
C) Neutral
D) No way to determine the charge on Plate 2

5) A proton and an electron are placed in an unform electric field. Which undergoes the greater acceleration?

A) proton
B) electron
C) Both accelerate equally.
D) Neither accelerates.

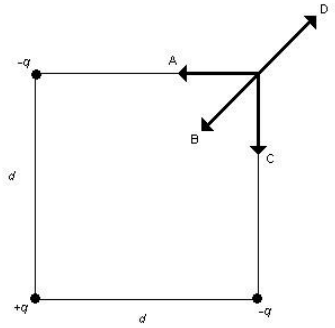
6) An electron is initially moving to the right when it enters a uniform electric field directed upwards, as shown in the figure. Which trajectory (**X**, **Y**, **Z**, or **W**) will the electron follow in the field?

A) trajectory **W**
B) trajectory **X**
C) trajectory **Y**
D) trajectory **Z**



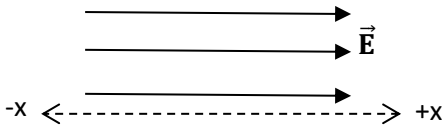
7) Three equal point charges of varying signs are placed on three of the corners of a square of side *d* as shown in the figure below. Which of the arrows shown represents the direction of the net electric field at the vacant corner of the square?

A) A
B) B
C) C
D) D
E) none of the above



8) A force of 6.0 N acts on a charge of 3.0 μC when it is placed in a uniform electric field. What is the magnitude of this electric field?

9) A proton is placed in an electric field of intensity 700 N/C. What is the magnitude and direction of the acceleration of this proton due to this field? mass of proton = 1.67×10^{-27} kg, $q_p = 1.6 \times 10^{-19}$ C



10) A small object with a 5.0-μC charge is accelerating horizontally on a friction-free surface at 0.0050 m/s^2 due only to an electric field. If the object has a mass of 2.0 g, what is the magnitude of the electric field?

11) A small 0.050-kg insulating sphere carries a negative charge of 60 μC and is hanging by a vertical silk thread from a fixed point in the ceiling. An external uniform vertical electric field is now applied. If the applied electric field has a magnitude of 3000 N/C and is directed downward, what is the tension in the silk thread? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)

12) A particle with a positive charge of 4.0 μC has a mass of 5.0×10^{-3} kg. What electric field directed upward will exactly balance the weight of the particle and in what direction (up or down)?

13) A Styrofoam ball of mass 0.120 g is placed in an electric field of 6000 N/C pointing downward. What charge must be placed on the ball for it to be suspended?

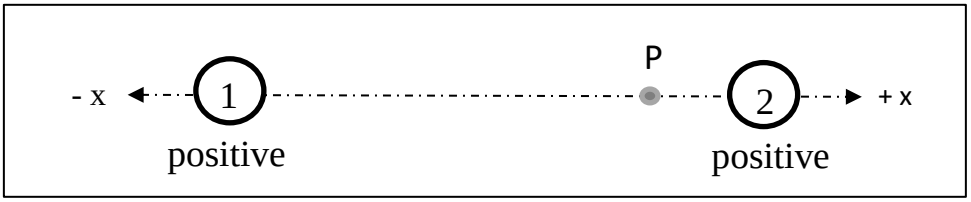
14) What are the magnitude and direction of the electric field at a distance of 1.5 m from a positive 50 nC charged sphere?

15) A small glass bead has been charged to 1.9 nC. What is the strength of the electric field 2.0 cm from the center of the bead? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)

16) The electric field at a point 2.8 cm from a small object points toward the object with a strength of 180,000 N/C. What is the object's charge q? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)

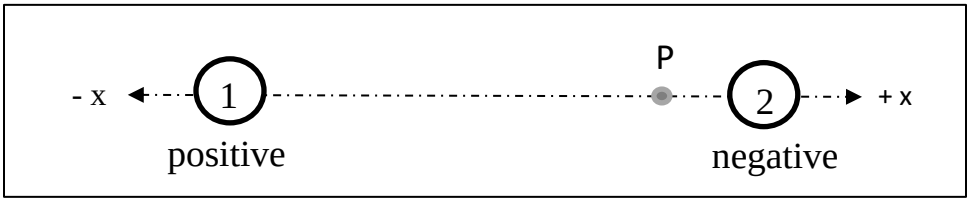
17) Two identically charged objects are shown below, what is the direction of the electric field at point P due to the two charged objects?

A) Positive x direction
B) Negative x direction
C) Zero electric field

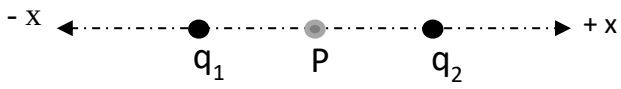


18) Two identically charged objects are shown below, what is the direction of the electric field at point P due to the two charged objects?

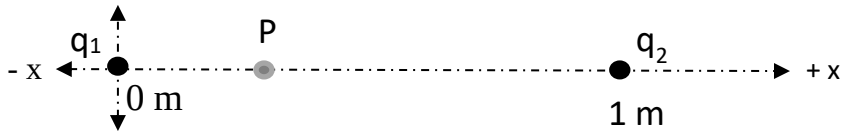
A) Positive x direction
B) Negative x direction
C) Zero electric field



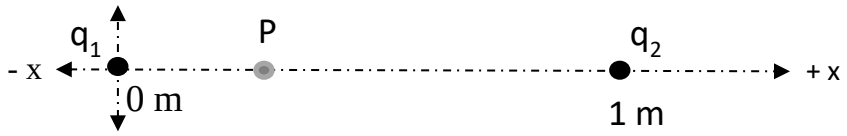
19) Two charged spheres with a net charge of $q_1 = 20\ \mu C$ positive and $q_2 = 8\ \mu C$ negative, are separated by a distance of 20 cm. What is the magnitude of electric field midway between these two charges at point P?



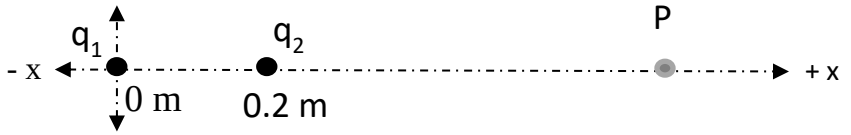
20) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ positive at the origin and $q_2 = 4\ \mu C$ negative and placed at the 1 m mark, what is the magnitude of the electric field at the 0.2 meter mark indicated by point P



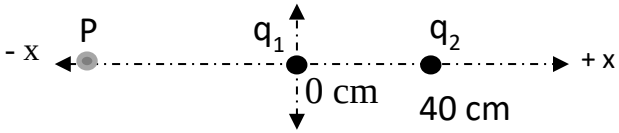
21) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ **negative** at the origin and $q_2 = 4\ \mu C$ negative and placed at the 1 m mark, what is the magnitude of the electric field at the 0.2 meter mark indicated by point P



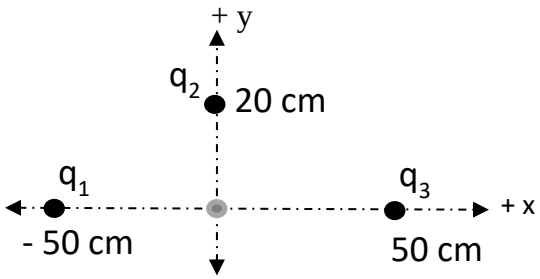
22) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ **negative** at the origin and $q_2 = 4\ \mu C$ positive and placed at the 0.2 m mark, what is the magnitude of the electric field at the 1 meter mark indicated by point P



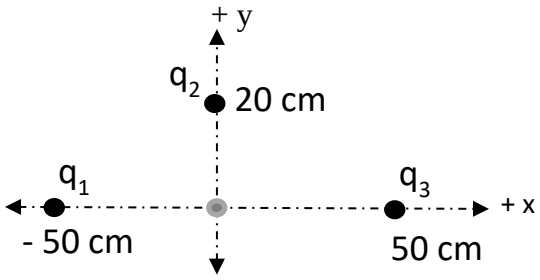
23) Two tiny particles having charges of $q_1 = 7$ μC positive and $q_2 = ?$ are placed along the *x*-axis. The positive 7 μC particle is at $x = 0$ cm, and q_2 is at $x = +40$ cm. If the net electric field strength is zero at point P located at – 3 m what must the charge on q_2 be and what type of charge (positive or negative)?



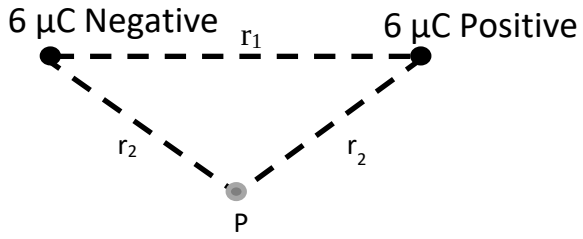
24) Three charged spheres are placed as shown, determine the magnitude and direction of the electric field at the origin of the coordinate system if $q_1 = 8$ μC positive, $q_2 = 6$ μC negative, and $q_3 = 8$ μC positive.



25) Three charged spheres are placed as shown, determine the magnitude and direction of the electric field at the origin of the coordinate system if $q_1 = 8$ μC positive, $q_2 = 6$ μC negative, and $q_3 = 8$ μC **negative**.



26) An electric dipole consists of two charged objects, if the net charge on the first object is a negative 6.00 μC and the second has a net charge of positive 6 μC, they are r_1 = 10 cm apart, as shown in the figure. Find the magnitude and direction of the electric field this dipole produces at point P, which is r_2 = 7.00 cm from each charge. ($k = 9 \times 10^9\ \mathrm{N \cdot m^2/C^2}$)



Answers

1) A
2) B
3) A
4) B
5) B
6) D
7) B
8) 2×10^{6} N/C
9) 6.7×10^{10} m/s^2
10) 2 N/C
11) 0.31 N
12) 12,250 N/C
13) 1.96×10^{-7} C
14) 200 N/C directed away from sphere
15) 42,750 N/C
16) 1.57×10^{-8} C
17) B
18) A
19) 2.52×10^{7} N/C
20) 5.06×10^{5} N/C
21) 3.94×10^{5} N/C in the negative x direction
22) 3.83×10^{4} N/C in the positive x direction
23) 9.0×10^{-6} C
24) 1.35×10^{6} N/C along positive y axis $\alpha = 90°$ and $\beta = 0°$
25) 1.47×10^{6} N/C $\alpha = 67°$ and $\beta = 23°$
26) 1.57×10^{7} N/C along negative x axis $\alpha = 180°$ and $\beta = 90°$